

Skills Summary

Interior and Exterior Angles of Polygons

Use this reference while completing your worksheet or when studying.

Skill 1 — The interior angle sum of a polygon

Rule: Diagonals drawn from one vertex cut an n -sided polygon into $n - 2$ triangles, and each triangle holds 180° : $S = (n - 2) \cdot 180^\circ$.

Read it forwards to get the sum from the sides, backwards to get the sides from the sum.

Example: Find the interior angle sum of an octagon.

Step 1. An octagon has 8 sides and I want the total of its angles — that is exactly what the interior-angle-sum rule gives, so no equation is needed.

Step 2. $S = (8 - 2) \cdot 180^\circ = 6 \cdot 180^\circ$

$$\Rightarrow S = 1080^\circ$$

Try it: Find the interior angle sum of a decagon (10 sides).

check: 1440°

Skill 2 — Finding one missing interior angle

Rule: The listed angles plus the unknown one must equal $(n - 2) \cdot 180^\circ$. Write that as an equation and solve for the unknown — do not average, and do not assume the polygon is regular.

Example: Three angles of a quadrilateral are 95° , 140° and 72° . Find the fourth.

Step 1. The four angles must total a fixed amount, and three of them are known — so this is an addition equation with one unknown, not a formula to evaluate.

Step 2. $(4 - 2) \cdot 180^\circ = 360^\circ$, so $95 + 140 + 72 + x = 360$, i.e. $307 + x = 360$

$$\Rightarrow x = 53^\circ$$

Try it: Three angles of a quadrilateral are 108° , 76° and 121° . Find the fourth.

check: 55°

Skill 3 — Exterior angles and the full turn

Rules: Walking once around any convex polygon turns you through one full circle, so the exterior angles always add to 360° — no matter how many sides. In a *regular* polygon they are all equal, so each one is $360^\circ/n$. At any single vertex, interior + exterior = 180° .

Example: Find each exterior angle of a regular polygon with 12 sides.

Step 1. The polygon is regular, so all its exterior angles are equal — share the one full turn out evenly instead of using the interior-angle rule.

Step 2. each exterior angle = $360^\circ/12$

$$\Rightarrow 30^\circ$$

Try it: Find each exterior angle of a regular polygon with 20 sides.

check: 18°

Skill 4 — From an angle back to the sides

Rule: $n = 360^\circ \div (\text{one exterior angle})$. If you are handed an *interior* angle instead, subtract it from 180° first to get the exterior angle, then divide.

Example: Each exterior angle of a regular polygon is 45° . How many sides?

Step 1. I know one exterior angle and want the count of equal pieces the full turn was cut into — that is a division, and it uses the exterior angle rather than the interior one.

Step 2. $n = 360^\circ / 45^\circ$

$$\Rightarrow n = 8$$

Try it: Each interior angle of a regular polygon is 150° . How many sides?

check: 12

(!) **Watch out:** $(n - 2) \cdot 180^\circ$ is the sum of *all* the interior angles. To get *one* interior angle of a regular polygon you must still divide by n . And the exterior-angle total is 360° for every polygon — it never depends on the number of sides.